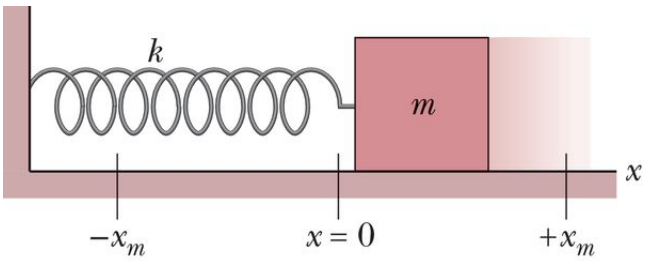
# Lesson 15A Simple Harmonic Motion

## Introduction to Simple Harmonic Motion

**Simple Harmonic Motion (SHM)** is a type of periodic motion in which an object oscillates back and forth in a regular, repeating pattern. It is characterized by motion that follows a sinusoidal function of time, such as a cosine or sine wave. A mass on a horizontal spring as shown below is a good example of a simple harmonic oscillator:




### Key Concepts

- **Frequency (f)** is the number of complete oscillations (cycles) per second. It is measured in **hertz (Hz)**, where

$$1\,\text{Hz} = 1\,\text{cycle/second}$$

- **Period (T)** is the time it takes to complete one full oscillation. It is related to frequency by:

$$T = \frac{1}{f}$$

- **Amplitude ($x_m$)** is the maximum displacement from equilibrium—how far the object moves in either direction.
- **Phase** describes the state of the motion at a given time. The full expression for position as a function of time in SHM is:

$$x(t) = x_m \cos(\omega t + \phi)$$

  - $\omega$ is the **angular frequency** (in radians per second)

- $\phi$ is the **phase constant**, which accounts for the initial conditions at $t = 0$

- The **angular frequency** $\omega$ is related to the frequency and period:

$$\omega = 2\pi f = \frac{2\pi}{T}$$

## Velocity and Acceleration

- **Velocity** is the time derivative of position:

$$v(t) = \frac{dx}{dt} = -\omega x_m \sin(\omega t + \phi)$$

  The **maximum velocity** (velocity amplitude) is:

$$v_m = \omega x_m$$

- **Acceleration** is the time derivative of velocity (second derivative of position):

$$a(t) = \frac{dv}{dt} = -\omega^2 x_m \cos(\omega t + \phi) = -\omega^2 x(t)$$

  The **acceleration amplitude** is:

$$a_m = \omega^2 x_m$$

  This equation shows that acceleration is directly proportional to displacement and directed toward the equilibrium point—this is the defining feature of SHM.

## Forces and Hooke's Law

According to Newton's second law:

$$F = ma = -m\omega^2 x$$

Comparing this to **Hooke's Law**, $F = -kx$, we find:

$$\omega = \sqrt{\frac{k}{m}}$$

From this, the **period** of a **linear simple harmonic oscillator** (like a mass on a spring) is:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

This confirms that a mass on a spring is a linear simple harmonic oscillator, where the restoring force is proportional to displacement.

## Energy in SHM

Consider a mass oscillating on a horizontal spring. The total mechanical energy is conserved and is a combination of:

- **Kinetic Energy**:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 x_m^2 \sin^2(\omega t + \phi)$$

- **Potential Energy (Spring)**:

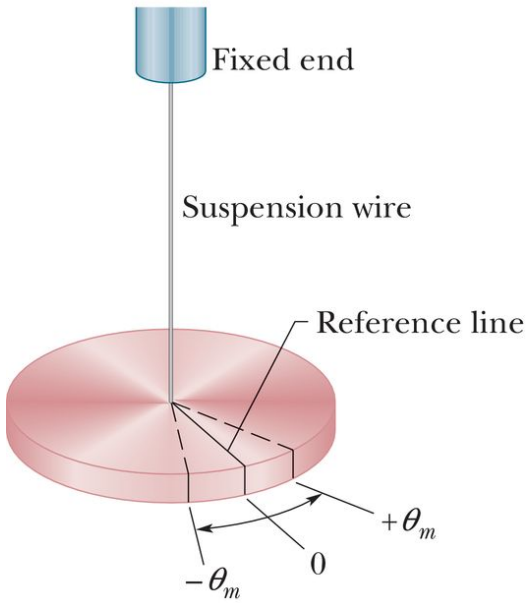
$$U = \frac{1}{2}kx^2 = \frac{1}{2}kx_m^2 \cos^2(\omega t + \phi)$$

- **Total Energy** (constant over time):

$$E = K + U = \frac{1}{2}kx_m^2$$

## Angular Simple Harmonic Oscillator

A **torsion pendulum** provides an example of angular SHM. It consists of a disk or rod suspended by a twisting wire as shown below:




The restoring torque is proportional to angular displacement:

$$\tau = -\kappa\theta$$

Where:

- $\tau$ is torque
- $\kappa$ is the **torsion constant**
- $\theta$ is angular displacement

The angular form of Newton's second law:

$$I\alpha = -\kappa\theta$$

Leads to:

$$\alpha = -\frac{\kappa}{I}\theta$$

This has the same form as linear SHM, with angular frequency:

$$\omega = \sqrt{\frac{\kappa}{I}}$$

The **period** of an **angular simple harmonic oscillator** is:

$$T = 2\pi\sqrt{\frac{I}{\kappa}}$$